

Garv Sharma

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EDUCATION

- ❖ Computer Science and Engineering (B.Tech + M.Tech)
National Institute of Technology, Kurukshetra (Haryana)
SGPA - 1st Sem- **8.92**, 2nd Sem-**8.97**, 3rd Sem - **9.1**
Aug 2024 - Aug 2029
- ❖ XII (ICSE) | Rishi Public School, Gurugram (Haryana)
92.4% | 2023

TECHNICAL SKILLS

Programming Languages - Java, C, C++, HTML, CSS, JavaScript, Python

Frameworks & Technologies - Spring Boot, REST APIs, Flutter, React.js

AI & Database Technologies - RAG Architectures, Vector Search, Local LLMs (Llama 3.2), MongoDB Atlas, SQL

Tools - Git, IntelliJ IDEA, VS Code, Eclipse, Docker, Postman

CS FUNDAMENTALS

Data Structures & Algorithms, Object-Oriented Programming (OOPs), Digital System Design, Computer Organization and Architecture, Operating Systems (OS), Database Management System

PROJECTS

- ❖ **Link Shortener** | Flutter, Spring Boot, MongoDB Atlas, Railway, Postman
 - Developed a full-stack, cross-platform link shortener application to efficiently generate, resolve and redirect short URLs.
 - Engineered backend REST APIs using Spring Boot and integrated MongoDB Atlas to manage and store original URLs and short codes.
 - Built the frontend interface using Flutter and systematically validated API endpoints using Postman prior to deployment on Railway.
- ❖ **Smart Cycle Lock (IoT, Embedded Systems)** | C++, Microcontroller (ESP-32), Wi-Fi Module, Sensors
 - Developed robust C++ firmware for an ESP-32 microcontroller to enable remote lock and unlock functionality over Wi-Fi.
 - Integrated hardware sensors and wireless modules to architect a responsive and secure IoT embedded system.
- ❖ **Smart Attendance System (SIH - 2025)** | Flutter, local_auth, permission_handler
 - Co-developed a responsive frontend user interface using Flutter to facilitate seamless and intuitive attendance tracking.
 - Implemented secure biometric authentication utilizing local_auth and permission_handler to ensure reliable user access control.
- ❖ **YouTube Semantic Search Engine** | React.js, Spring Boot, MongoDB Atlas (Vector Search), Llama 3.2, YouTube Data API
 - Architected a full-stack semantic search application that retrieves and ranks YouTube videos based on natural language user intent rather than exact keyword matches.
 - Engineered a Retrieval-Augmented Generation (RAG) pipeline utilizing a locally deployed Llama 3.2 model to broaden search intent and intelligently rank results.
 - Integrated the YouTube Data API for data ingestion and leveraged MongoDB Atlas Vector Search for efficient high-dimensional vector storage and retrieval.
- ❖ **Team Finder** | React.js, Spring Boot, MongoDB Atlas, Postman, Docker, Hugging Face, Vercel, Google/GitHub OAuth
 - Developed a full-stack web application designed to help users seamlessly find and connect with teammates for collaborative projects and hackathons.
 - Engineered a robust REST API using Spring Boot to power a student collaboration platform, efficiently serving data to a React.js frontend.
 - Implemented secure user authentication by integrating Google OAuth and GitHub Login, ensuring safe and seamless access.
 - Integrated MongoDB Atlas for scalable data management and implemented a lightweight chat feature to facilitate initial user communication.
 - Optimized backend performance and reduced database query latency by implementing caching mechanisms for high-traffic endpoints.
 - Optimized the platform for search engines (SEO), implementing best practices that achieved top search result rankings and significantly improved organic discoverability.

POSITION OF RESPONSIBILITY

❖ Team Leader - Smart Cycle Lock Project

Led a 5-member team, coordinated task distribution, tracked progress, and ensured successful on - time project delivery.

ACHIEVEMENTS

- ❖ Secured **AIR- 880 in ICPC-2025 Preliminary contest** as part of team '**Survival Trio**' (solved 3 out of 6 problems).
- ❖ Secured All India Rank **20,961 (Top 1.5%)** in JEE Mains 2024 among 1.4 Million candidates.
- ❖ Codechef - 1410 (2 star)
- ❖ Codeforces - 1095 (max)
- ❖ Leetcode - 1642 (Top 18.74%)
- ❖ Solved 1000+ Data Structures and Algorithm Problems.

CODING PROFILE

- ❖ Codeolio Portfolio - [Garv Sharma | Codolio](#)
- ❖ Leetcode - [garvsharma09 - LeetCode](#)
- ❖ GeeksforGeeks - [GeeksforGeeks - Garv Sharma](#)
- ❖ Codeforces - <https://codeforces.com/profile/garvroc9>
- ❖ Codechef - [suave_shop_50 | CodeChef User Profile for Garv Sharma | CodeChef](#)